



SHAH & ANCHOR KUTCHHI ENGINEERING COLLEGE
(An Autonomous Institute Affiliated to University of Mumbai)
MUMBAI - 400 088, MAHARASHTRA (INDIA)
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Subject :- Advanced Database Management System [ADBMS]
Self Learning Assessment Report [SLA]

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Topic: Big Data

Chapter 1: Title

Big Data: Characteristics (5 V's), Technologies, Processing Frameworks (Hadoop, MapReduce, Spark), and Architectures

Chapter 2: Key Syllabus Points

- Understanding the **concept and characteristics of Big Data**
- Study of **5 V's of Big Data**: Volume, Velocity, Variety, Veracity, Value
- Understanding **Big Data Technologies**:
 - Hadoop Ecosystem (HDFS, YARN)
 - MapReduce programming model
 - Apache Spark
 - NoSQL databases
- Exploring **Big Data Architectures**: Batch Processing, Real-time Processing, Lambda, Kappa

Chapter 3: Introduction

Big Data refers to **extremely large datasets that cannot be processed using traditional database tools**. It encompasses data that is too voluminous, fast-moving, or complex for conventional systems.

Big Data is characterized by the **5 V's**: Volume (scale), Velocity (speed), Variety (types), Veracity (quality), and Value (insights). Technologies like Hadoop and Spark enable organizations to process petabytes of data efficiently, transforming raw information into actionable insights for business intelligence and strategic decision-making.

Chapter 4: SDG Goal & Target

This project aligns with **SDG Goal 9: Industry, Innovation, and Infrastructure**.

Relevant Target:

- **Goal 9.5**: Enhance scientific research and upgrade technological capabilities through advanced data analytics.

Big Data supports this SDG by enabling **data-driven innovation**, **efficient decision-making**, and **evidence-based policy making** in industries and governments.

Chapter 5: Problem Identification

Modern organizations face critical challenges with data management:

- **Exponential data growth:** Generation of massive amounts of data daily from IoT devices, social media, and transactions.
- **Traditional database limitations:** Conventional systems cannot scale to handle petabytes of data efficiently.
- **Real-time processing needs:** Requirement for instant insights from streaming data.
- **Data variety challenges:** Handling structured, semi-structured, and unstructured data from multiple sources.

Traditional databases focus on **structured data and transactions**, not on scale and variety. Hence, there is a need for **Big Data technologies** that enable distributed storage and parallel processing.

Chapter 6: Background Work

Before Big Data technologies, organizations relied on traditional relational databases that struggled with web-scale data.

Evolution:

1. **2003-2006:** Google published papers on **MapReduce** and distributed computing concepts.
2. **2006:** **Apache Hadoop** was created as open-source distributed processing framework.
3. **2009:** **NoSQL databases** gained popularity for handling unstructured data.
4. **2014:** **Apache Spark** emerged, offering in-memory processing much faster than MapReduce.
5. **Present:** Integration with cloud-based Big Data services and real-time streaming frameworks.

Chapter 7: Objectives

- To understand the **fundamental concepts** of Big Data and the 5 V's.
- To explore **Big Data processing frameworks** including Hadoop, MapReduce, and Spark.
- To study **Big Data architectures** for batch and real-time processing.
- To demonstrate **practical applications** of Big Data in real-world scenarios.
- To enhance understanding of **distributed computing and scalable data processing**.
- To enhance understanding of **data warehousing and business intelligence concepts**.